Digital Payment Performance Analytics Report – UPI vs IMPS

Introduction:

Digital payments have become an important part of India's financial system, changing the way people and businesses transfer money. Among the major platforms, UPI and IMPS play a key role. UPI is widely used for everyday small-value transactions, while IMPS is trusted for high-value transfers. This project uses Python to study their monthly performance from 2016 to 2026, comparing transaction volume, value, and bank participation. The analysis helps to understand growth trends, usage patterns, and the complementary role of UPI and IMPS in strengthening India's digital payment ecosystem.

1. Project Overview:

This project focuses on analyzing UPI and IMPS transaction datasets using Python to extract meaningful insights. The analysis includes data cleaning, exploratory data analysis (EDA), statistical analysis, and visualization to understand growth patterns, seasonal trends, and comparative performance. The project aims to transform raw transaction data into useful information that supports better decision-making in India's digital payment ecosystem.

2. Data Source:

Dataset Name: UPI & IMPS Transaction Statistics

Source: NPCI Official Product Statistics

File Type: Excel (monthly data)

Timeline: April 2016 – March 2026

Attributes: Transaction Volume (Mn.), Transaction Value (₹ Cr.), Average Transaction Value, Bank Participation

3. Problem Statement:

The rapid growth of digital payments has transformed financial transactions in India. UPI and IMPS are two major platforms, but their performance and growth patterns differ. Understanding these differences is crucial for banks and policymakers. However, limited comparative analysis has been conducted using official transaction data. This project uses Python to analyze and compare UPI and IMPS transactions, generating insights to guide adoption and strengthen India's digital payment ecosystem.

4. Objectives:

- To analyze monthly transaction performance of UPI and IMPS (2016–2026) using Python.
- To compare transaction volume and value to evaluate growth trends.
- To examine year-wise and quarter-wise performance for seasonal trends.
- To study comparative growth, market performance, and transaction behavior through statistical and graphical analysis.

- To provide data-driven insights and recommendations on adoption and performance of digital payment systems in India.

5. Attribute (Column / Feature) Details:

Attribute Name	Description
Year	Year of transaction statistics
Month	Month of transaction statistics
UPI Volume (Mn.)	Total UPI transaction volume
UPI Value (₹ Cr.)	Total UPI transaction value
IMPS Volume (Mn.)	Total IMPS transaction volume
IMPS Value (₹ Cr.)	Total IMPS transaction value
Average Transaction Value	Value ÷ Volume
Number of Banks	Banks live on UPI / IMPS

6. Tools & Technologies:

- Python
- Pandas, NumPy
- Matplotlib, Seaborn, Plotly
- Google Colab

7. Data Pre-Processing and Feature Engineering:

Data Cleaning Steps:

1. Checked data types of all columns
2. Identified missing values
3. Removed duplicate records
4. Converted transaction values to numeric
5. Standardized the month column to datetime

Feature Engineering:

1. Financial year mapping (April–March)
2. Transaction Growth % calculation
3. Average Transaction Value derived
4. Quarter feature extracted

8. Analysis and Visualizations:

The dataset was analyzed using EDA techniques to understand transaction growth, adoption, and comparative performance.

9. Exploratory Data Analysis (EDA):

- Verified dataset structure
- Statistical summary of transaction columns

- Null value and duplicate checks
- Distribution analysis of transaction volume, value, and bank participation

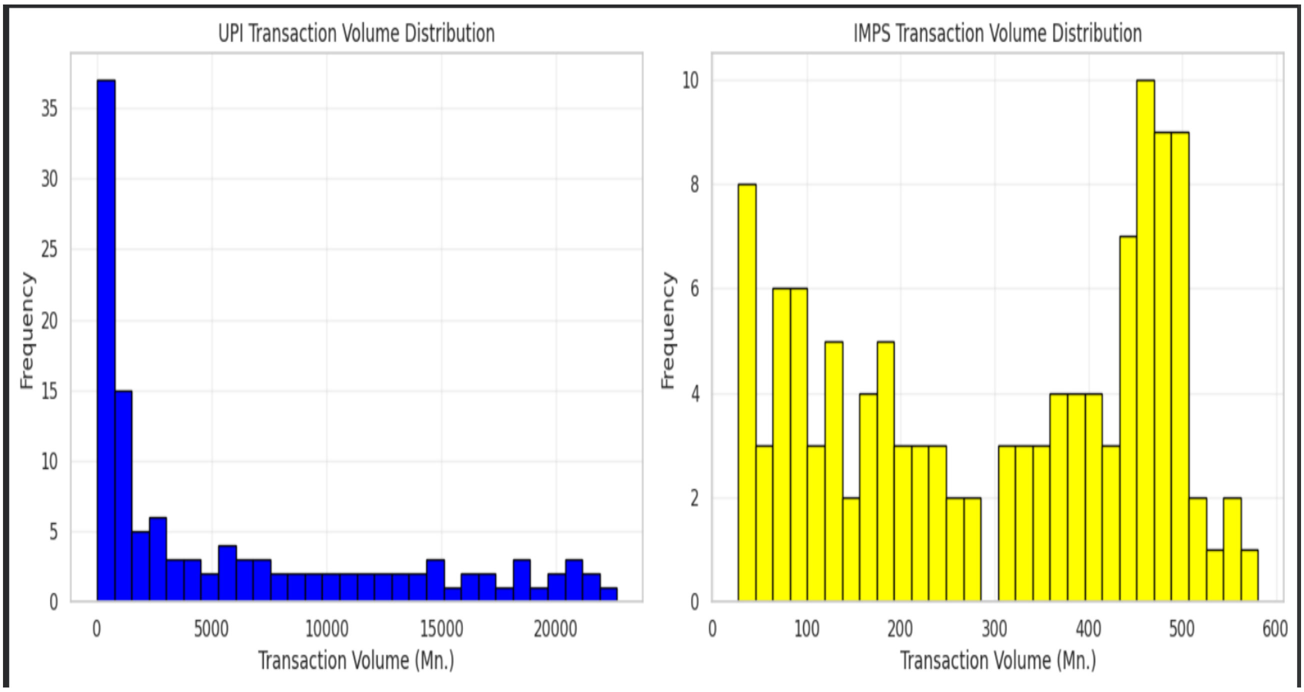
10. Statistical Analysis:

- UPI mean volume = 6,079.21 Mn. vs IMPS = 300.10 Mn. → UPI shows stronger adoption.
- UPI mean value = ₹8,92,022 Cr. vs IMPS = ₹3,26,927 Cr. → UPI handles larger overall value.
- Average transaction value → IMPS higher (₹1,025.64) vs UPI (₹175.74).
- Variance & Std Dev → UPI much higher, showing greater volatility in growth.
- Skewness → UPI positive (0.95), IMPS slightly negative (-0.21).
- Kurtosis → Both are negative (UPI -0.45, IMPS -1.46), flatter than a normal distribution.

Overall Insight: UPI dominates in volume & total value, while IMPS leads in average transaction value.

11. Univariate Analysis:

1.Transaction Volume Distribution → UPI avg 6,079.21 Mn., range 0–22,641.11 Mn.; IMPSavg 300.10 Mn., range 26.78–580.64 Mn.



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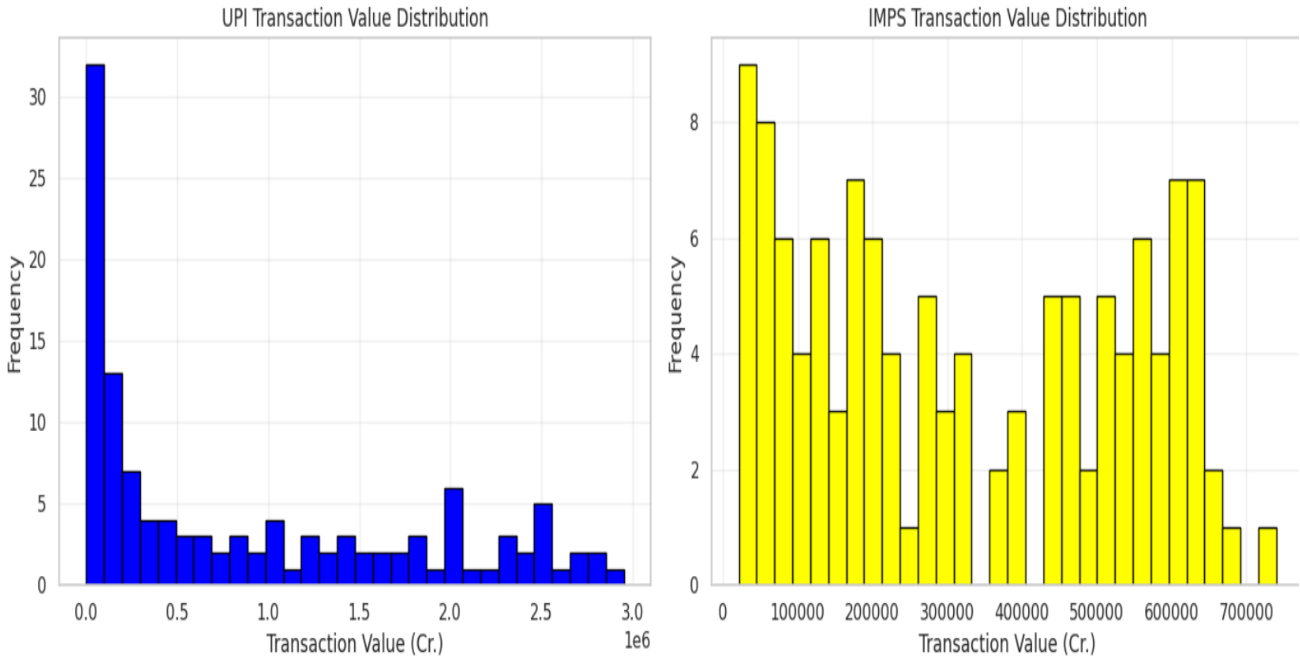
2.Transaction Value Distribution → UPI avg ₹8,92,022.30 Cr., range ₹0–₹29,52,542.05 Cr.; IMPS avg ₹3,26,926.98 Cr., range ₹21,044.30–₹7,40,176.19 Cr.



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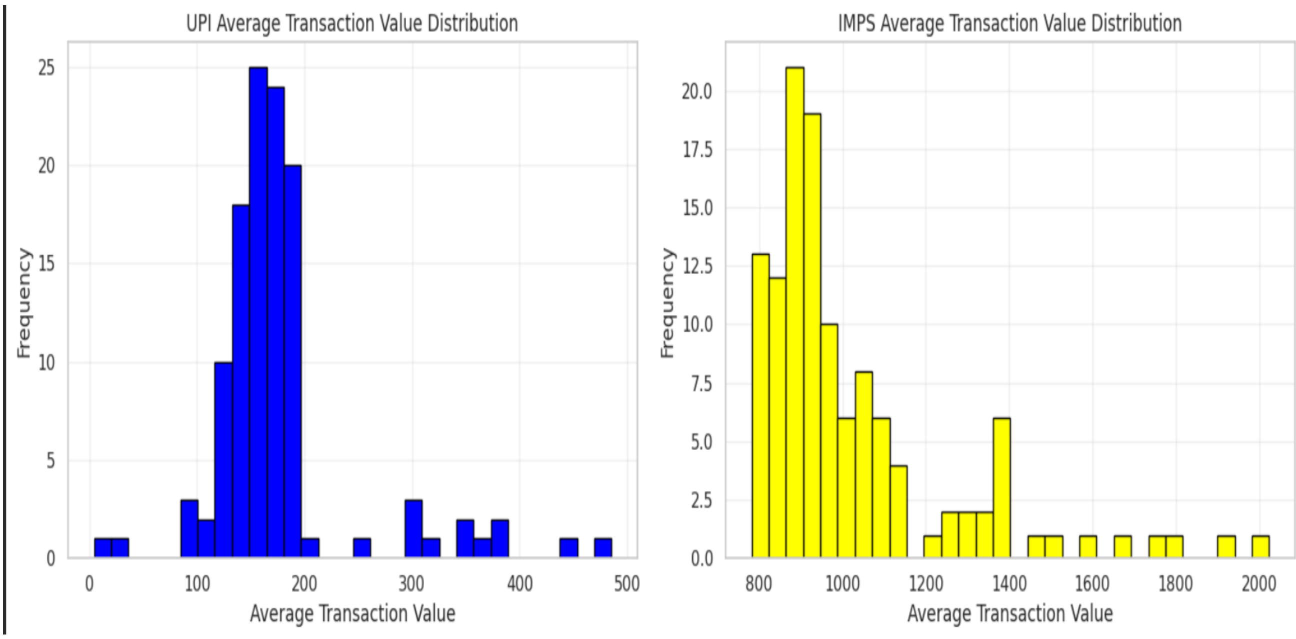
3.Average Transaction Value Distribution → UPI avg ₹171.34, range ₹0–₹485.70; IMPS avg ₹1,025.64, range ₹781.55–₹2,024.11.



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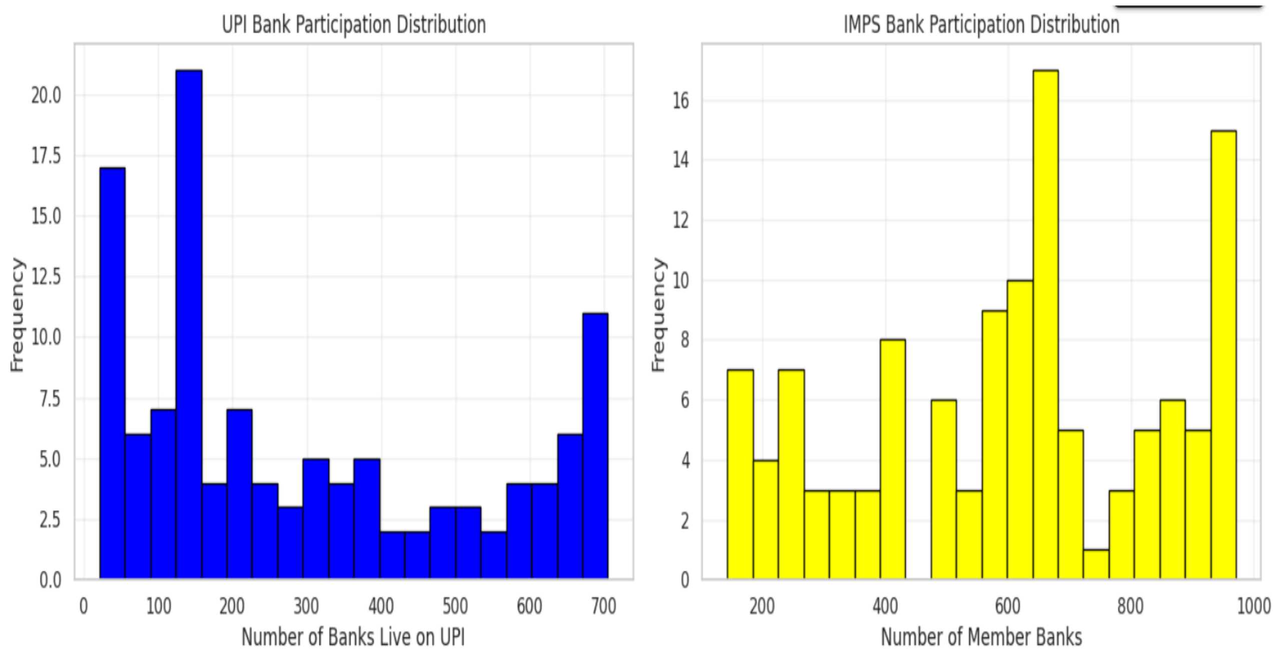
4.Bank Participation Distribution → UPI banks 21→705; IMPS banks 143→971.



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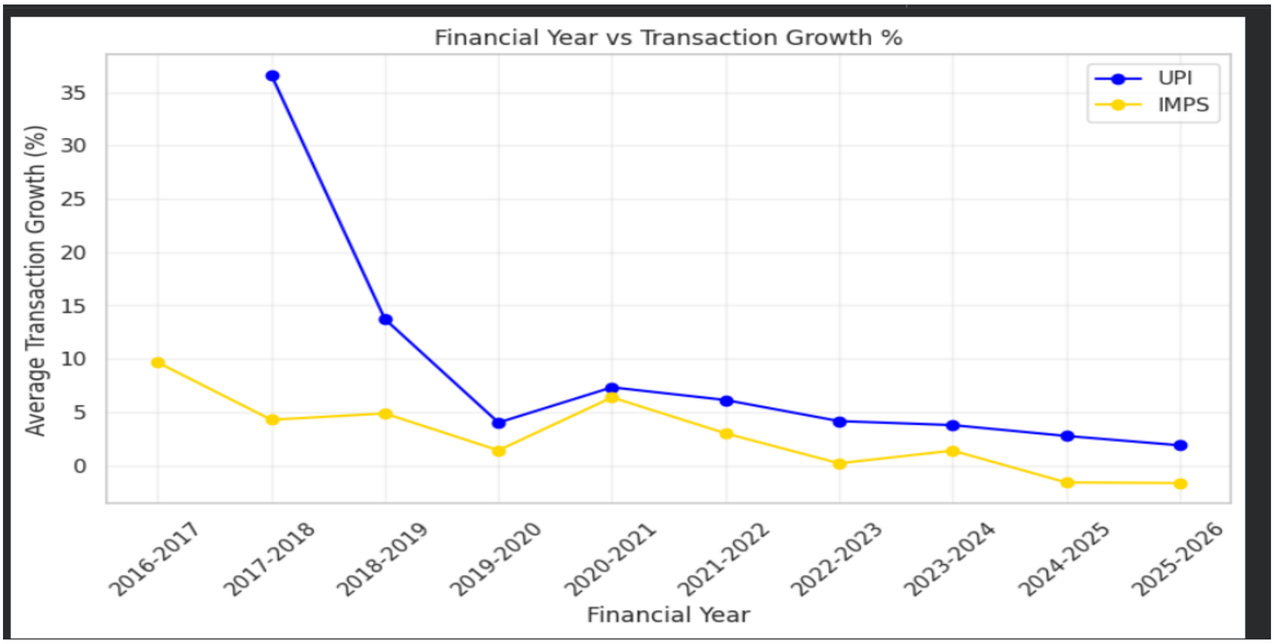
12. Bivariate Analysis:

1.Financial Year vs Growth % → UPI peak ≈37% (2017–18), decline to ≈2–3% (2025–26); IMPS ≈10% (2016–17), decline to ≈0–1% (2025–26).



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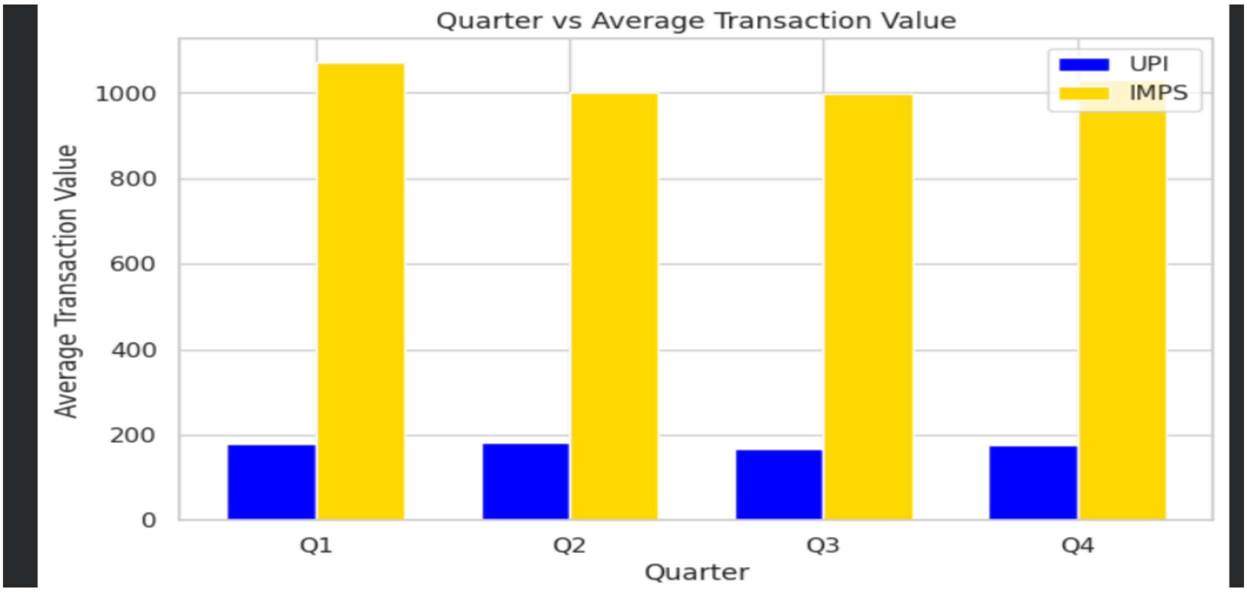
2.Quarter vs Average Transaction Value → UPI ₹170–₹180 across Q1–Q4; IMPS ₹1000–₹1100, gap ₹800–₹900.



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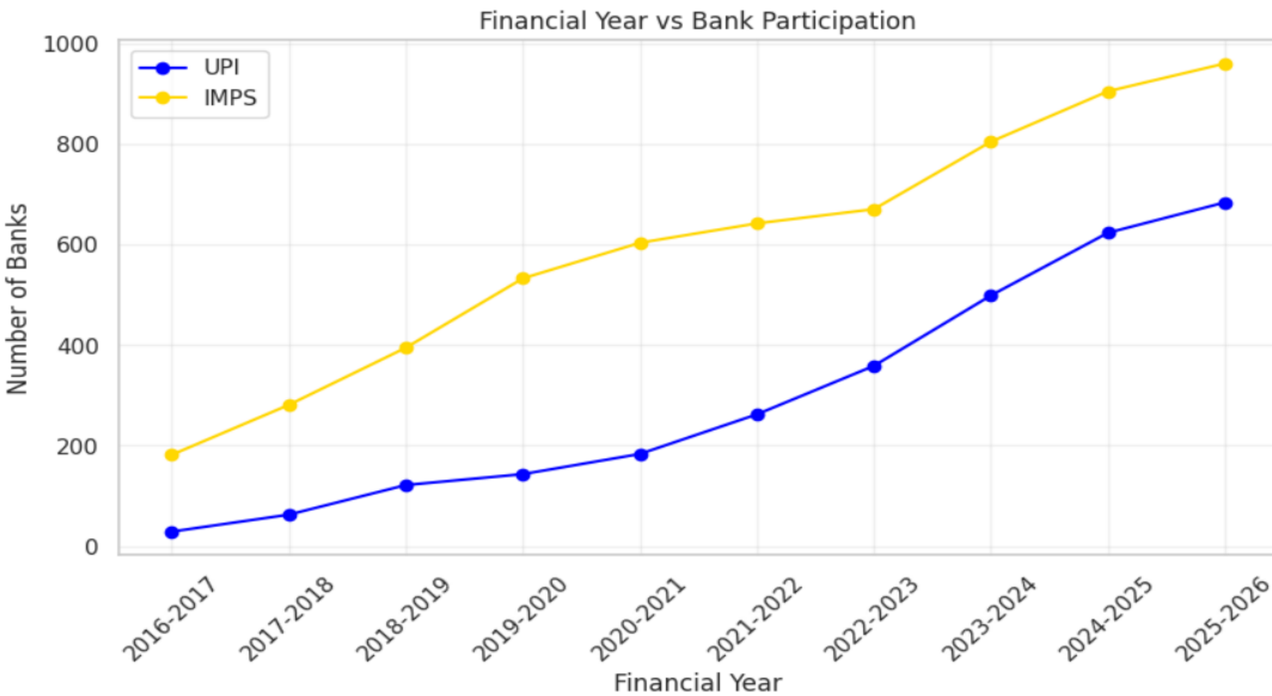
3.Financial Year vs Bank Participation → UPI ≈30→700 banks; IMPS ≈180→950 banks.



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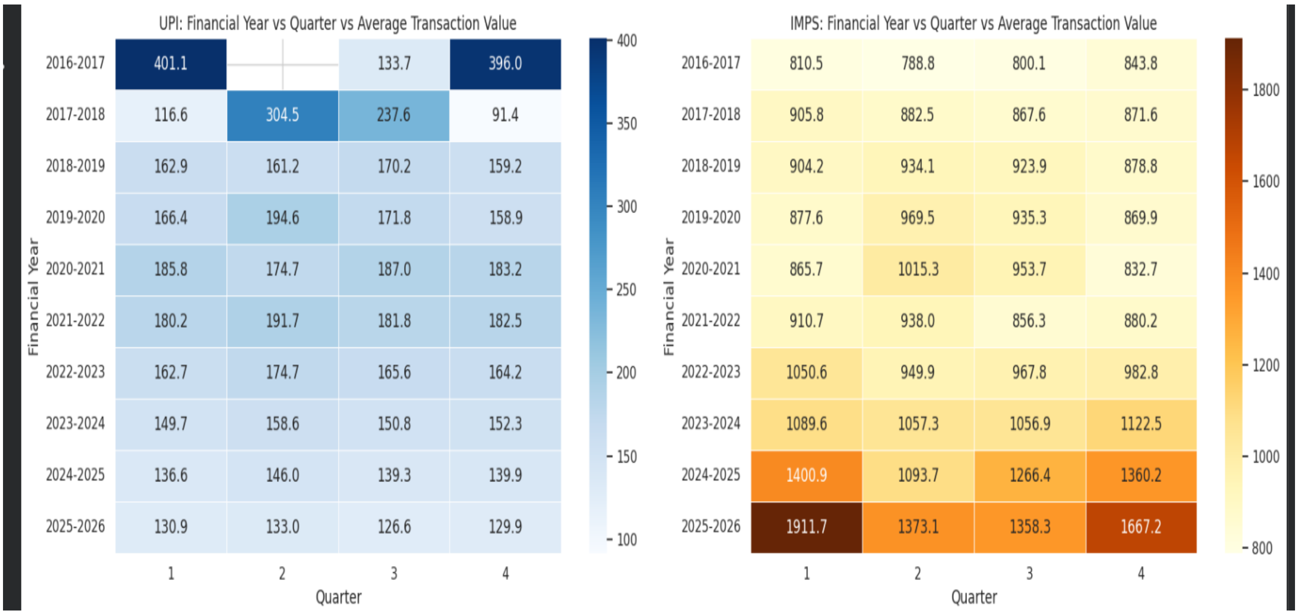
13. Multivariate Analysis:

1.Year × Quarter × Avg Transaction Value (Heatmap) → UPI values ₹130–₹400, stabilize near ₹170–₹180; IMPS values ₹800–₹1900, rising to ₹1900 (2025–26).



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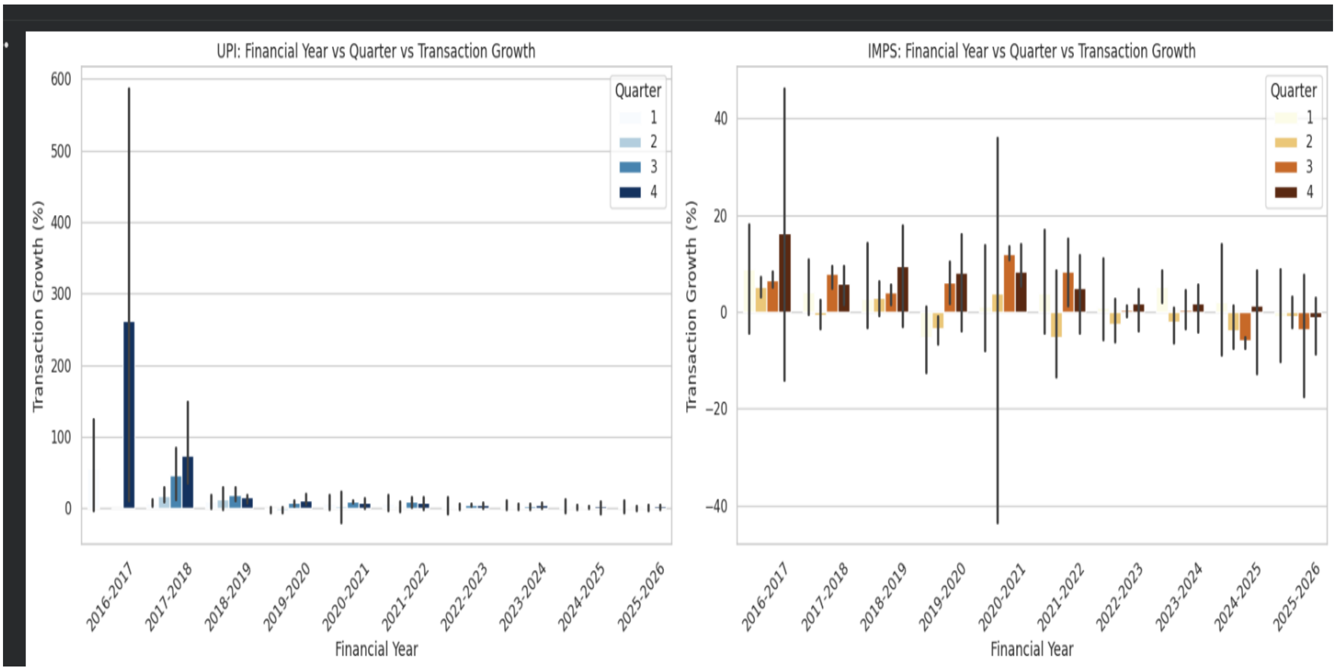
2.Year × Quarter × Growth % (Bar plot) → UPI spikes 500%+ (2016–18), stabilizes < 50% later; IMPS fluctuates -40% to +40%, pandemic shocks.



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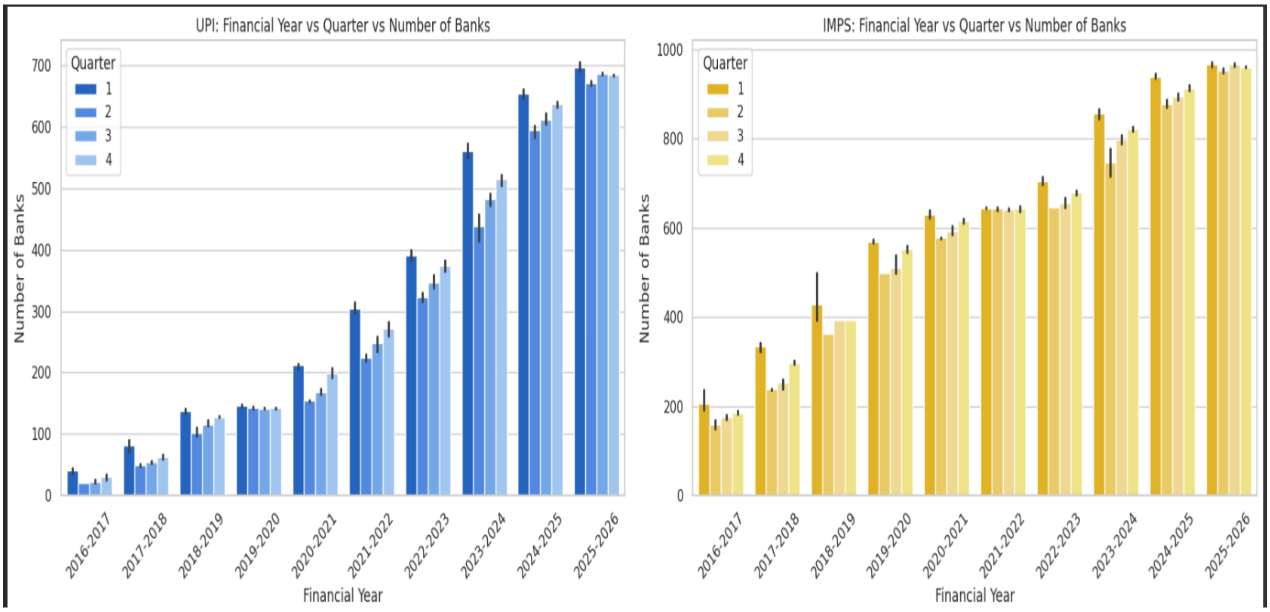
3.Year × Quarter × Bank Participation (Bar plot) → UPI ≈30→700 banks; IMPS ≈180→950 banks; UPI steeper growth, IMPS broader coverage.



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14. Summary of Findings:

- UPI Mean Transaction Volume: 6,079.21 Mn.
- IMPS Mean Transaction Volume: 300.10 Mn.
- UPI Mean Transaction Value: ₹8,92,022.31 Cr.
- IMPS Mean Transaction Value: ₹3,26,926.98 Cr.
- Average Transaction Value: UPI ≈ ₹175.74, IMPS ≈ ₹1,025.64
- Bank Participation: UPI 21 → 705 banks, IMPS 143 → 971 banks Overall Insight: UPI dominates in volume & total value, IMPS dominates in per-transaction value & coverage.

15. Types of Analysis:

1. Descriptive Analysis → UPI higher in volume & value, IMPS higher in average transaction value, both show rising bank participation.
2. Diagnostic Analysis → UPI's rapid rise driven by mobile adoption; IMPS stable due to legacy trust; pandemic boosted UPI usage.
3. Predictive Analysis → UPI expected to expand in volume, IMPS stable with high-value transfers, UPI values remain low, IMPS values may rise.
4. Prescriptive Analysis → Expand UPI in rural areas, strengthen IMPS for corporate transfers, enhance infrastructure & security, integrate both platforms.

16. Future Enhancements & Suggestions:

- Build a Power BI dashboard comparing UPI vs IMPS.
- Develop ML models to forecast adoption & growth.
- Perform time-series forecasting for values & bank participation.
- Integrate real-time payment data for live monitoring.
- Create a Streamlit web app for interactive visualization.
- Analyze regional adoption differences.

Conclusion:

The comparative study of UPI and IMPS transactions from 2016 to 2026 clearly shows their complementary roles in India's digital payment ecosystem. UPI dominates in transaction volume (6,079.21 Mn.) and overall transaction value (₹8,92,022.31 Cr.), reflecting mass consumer adoption and mobile-driven integration. In contrast, IMPS records a smaller volume (300.10 Mn.) and value (₹3,26,926.98 Cr.) but maintains a much higher average transaction value (₹1,025.64), highlighting its strength in corporate and interbank transfers. Bank participation expanded significantly for both platforms, with UPI growing from 21 to 705 banks and IMPS from 143 to 971 banks, confirming strong institutional support. Growth trends reveal UPI's steep early adoption curve stabilizing in later years, while IMPS maintained steady expansion, reflecting maturity and reliability. Overall, UPI powers everyday retail payments through high-volume, low-value transactions, while IMPS secures high-value transfers, together strengthening India's digital payment backbone.